

An Empirical Comparison of Logistic Regression, Linear SVM, and Random Forest on Three UCI Datasets

Abstract

In this project, three UCI Datasets: Heart Disease, Wine Quality, and Adult will be used to conduct an empirical comparison of three supervised learning algorithms: logistic regression, linear support vector machines, and random forests. Using the empirical-comparison paper from Caruana and Niculescu-Mizil as a guide, this project will evaluate how standard classifiers will behave throughout datasets and training sizes with a consistent experimental protocol. Following the project guideline, we focus on two-class classification and evaluate performance using classification accuracy. With each dataset, we perform stratified train/test splits using three training fractions: 20%, 50%, 80% and repeat each setting across three trials. The hyperparameters are selected using 5-fold stratified cross-validation on the training set using grid search. We extracted the mean and standard deviation of training, cross-validation, and test accuracy across the three trials, and the results show that Random Forest achieves the best performance on Wine Quality and Adult across all training fractions, while Logistic Regression performs better on Heart Disease. Throughout the different datasets, we found that increasing training size will most likely improve test accuracy, although the magnitude of improvement depends on the dataset characteristics and model capacity.

Introduction

When training a classifier, it is ideal to find the best algorithm for a new dataset. The empirical study by Caruana and Niculescu-Mizil found that modern ensemble methods and margin-based methods often perform strongly on average, but also show substantial variability across problems. The purpose of this project is to create a small version of the empirical comparison using a consistent experimental protocol. In which we compare Logistic Regression, Linear SVM, and Random Forest. And use the three models on three datasets with different sizes and feature characteristics: Heart Disease (small-sized data set with mostly numeric clinical attributes), Wine Quality (medium-sized data set with numeric chemical measurements, and is converted to a binary label using a threshold), and Adult (large-sized dataset with mixed numeric and categorical features, and is converted to binary income label). This comparison is designed to help answer two questions: Which classifier performs best for a dataset under each train/test split? Does performance improve as the amount of training data increases (20%, 50%, 80%)?

Data

In the project, we are using the following datasets and converting them into two-class classification problems:

Dataset	Total	Features	Positive rate
Heart Disease	303	13	0.459
Wine Quality	6497	12	0.197
Adult	48842	14	0.239

In Heart Disease, the standard target indicates disease presence vs absence. In Wine Quality, we binarized the quality score: positive if the quality is greater than or equal to 7, and negative otherwise. In Adult, if the income is greater than \$50k, then it will be positive, or else it will be negative. In the project, all models were implemented using scikit-learn pipelines, ensuring consistent preprocessing.

Method

Models:

1. Logistic Regression: we used a L_2 -regularized logistic regression model. With key hyperparameter: Regularization Strength $C \in \{0.01, 0.1, 1, 10\}$ (larger C = weaker regularization)
2. Linear SVM: we used a linear SVM classifier with the Key hyperparameter: $C \in \{0.01, 0.1, 1, 10\}$
3. Random Forest: We trained random forests with $n_estimators = 300$, with key hyperparameters:
 - a. $max_depth \in \{None, 10, 20\}$
 - b. $max_features \in \{"sqrt", "log2"\}$
 - c. $min_samples_leaf \in \{1, 5\}$

For each dataset and each training fraction $f \in \{0.2, 0.5, 0.8\}$, we generated a stratified train/test split with test size $1-f$, and repeated this procedure 3 times. On each training set, we selected hyperparameters using GridSearchCV with 5-fold StratifiedKFold cross-validation. And recorded the training accuracy, validation accuracy, and test accuracy.

Experiments and Results

```
=== Test accuracy (mean ± std), train_frac=0.2 ===
```

dataset	AdultIncome	HeartDisease	WineQuality
classifier			
LinSVM	0.850 ± 0.000	0.801 ± 0.031	0.817 ± 0.002
LogReg	0.852 ± 0.000	0.835 ± 0.007	0.815 ± 0.002
RandForest	0.861 ± 0.002	0.794 ± 0.008	0.847 ± 0.003

```
=== Test accuracy (mean ± std), train_frac=0.5 ===
```

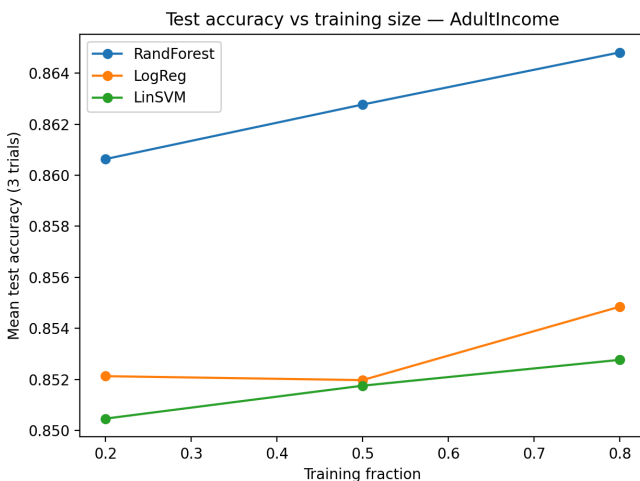
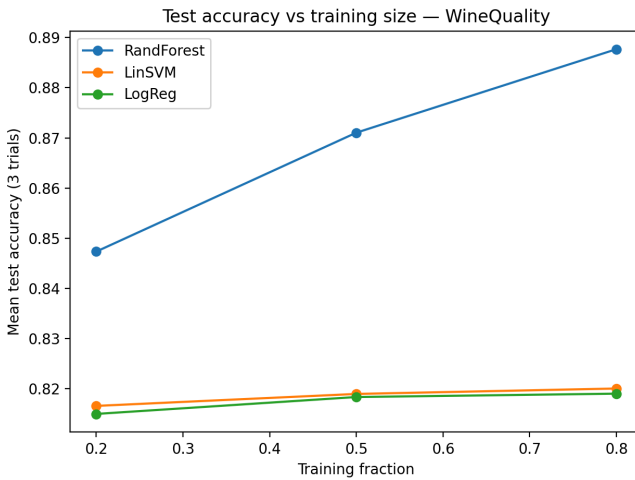
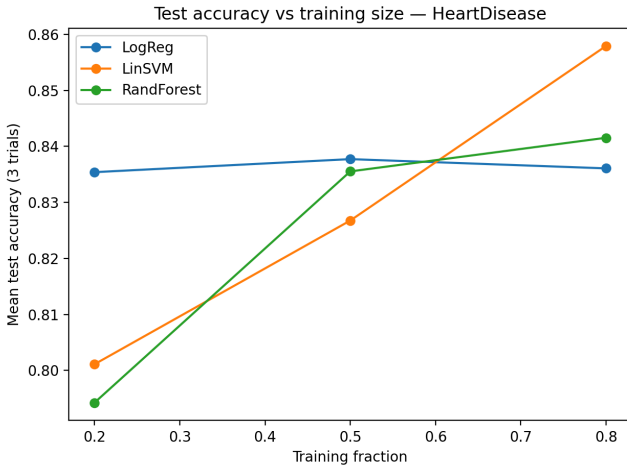
dataset	AdultIncome	HeartDisease	WineQuality
classifier			
LinSVM	0.852 ± 0.002	0.827 ± 0.023	0.819 ± 0.003
LogReg	0.852 ± 0.002	0.838 ± 0.025	0.818 ± 0.001
RandForest	0.863 ± 0.002	0.836 ± 0.017	0.871 ± 0.004

```
=== Test accuracy (mean ± std), train_frac=0.8 ===
```

dataset	AdultIncome	HeartDisease	WineQuality
classifier			
LinSVM	0.853 ± 0.002	0.858 ± 0.034	0.820 ± 0.002
LogReg	0.855 ± 0.003	0.836 ± 0.028	0.819 ± 0.005
RandForest	0.865 ± 0.002	0.842 ± 0.050	0.888 ± 0.003

The tables show the mean ± standard deviation test accuracy over three trials for each dataset and training fraction. In which we found:

- Adult Income:
 - Random forest has the best accuracy across all training fractions with small and consistent gains as training size increases.
 - Logistic regression and linear SVM are good methods, but are still a bit lower.
- Wine Quality:
 - Random forest improves substantially with more training data and performs better than linear models, which shows that nonlinear feature interactions are important for this task.
- Heart Disease:
 - We observe that Logistic Regression performs well with smaller training fractions. However, Linear SVM becomes the best method at the largest training fraction. The results are also noisier due to the small dataset size.



From the plots, we can see the comparison in test accuracy and training size. Throughout the different datasets, we can see that increasing the training fraction will most likely improve the test accuracy. We see the biggest improvement for Random Forest on Wine Quality, which shows that additional training data helps the ensemble to capture more complex patterns. On the other hand, Adult income improves more modestly with training size, which shows that we

might approach a performance ceiling for these models under the chosen feature representation. We also observe that Heart Disease exhibits the highest variance across trials, consistent with small-sample instability. Overall, the best performing method changes with training fraction, which proves the importance of repeated trials and caution in evaluation.

Training accuracy is used as a sanity check; Random Forests typically achieve higher training accuracy than linear models, consistent with their flexibility and tendency to overfit without regularization. When training accuracy is low for models, this may indicate preprocessing issues, label errors, or overly restrictive hyperparameters. Throughout the experiments, training accuracy patterns seem to be reasonable and follow the expectations for model capacity.

Conclusion

In conclusion, this project compared Logistic Regression, Linear SVM, and Random Forest on three UCI datasets using a consistent pipeline and cross-validated hyperparameter tuning. The results show that Random Forest performs strongly across diverse datasets and improves with increased training data, especially on Wine Quality, where nonlinear interactions are important. On the other hand, linear models still perform well on Adult and Heart Disease datasets, and particularly when the training data is limited, but present a smaller gain with additional data compared to Random Forest on more complex tasks. The Heart Disease dataset shows a greater instability across the experiments because of its small size, this shows the importance of repeated evaluation and reporting variability. With the results, we can prove that ensemble methods will most likely show strong performance, while simpler linear models can be more effective depending on the structure of the dataset.

References

- Caruana, R., & Niculescu-Mizil, A. (2006). *An Empirical Comparison of Supervised Learning Algorithms*. ICML.
- UCI Machine Learning Repository: Heart Disease, Wine Quality, Adult datasets.